Establishing the Need for Multimedia to Support Indigenous Knowledge Systems in South African Life Sciences Education: A Systematic Literature Review (2000-2026)

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Abstract

The integration of Indigenous Knowledge Systems (IKS) into South African Life Sciences education has gained increasing attention as part of ongoing curriculum transformation aimed at promoting culturally responsive and inclusive science teaching. Although the Curriculum and Assessment Policy Statement (CAPS) recognizes the importance of Indigenous Knowledge, research indicates that teachers continue to experience challenges related to instructional resources, pedagogical approaches, and professional preparation. This literature review examines the need for multimedia to support the integration of IKS in South African Life Sciences education. The review synthesizes literature on Indigenous Knowledge integration, multimedia in science education, multimedia learning theories, and instructional design principles to identify current challenges and opportunities for enhancing teaching and learning. The literature suggests that multimedia can improve learner engagement, facilitate conceptual understanding, support culturally responsive teaching, and provide authentic representations of Indigenous Knowledge through the use of text, images, audio, video, animations, and interactive learning activities. The review further demonstrates that constructivism, situated cognition, metacognition, Cognitive Load Theory, and Mayer's Cognitive Theory of Multimedia Learning provide a strong theoretical foundation for the design of multimedia resources that promote meaningful learning. Overall, the literature establishes a clear need for curriculum-aligned multimedia resources that support teachers in integrating Indigenous Knowledge into Life Sciences classrooms while enhancing learner engagement and promoting meaningful connections between Indigenous and scientific knowledge. The review concludes that the development of evidence-based multimedia resources has the potential to strengthen curriculum implementation and contribute to more inclusive and contextually relevant science education in South Africa.

Keywords: Indigenous Knowledge Systems, multimedia learning, Life Sciences education, science education, instructional design, South Africa, multimedia, culturally responsive teaching.

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1. Introduction

Science education plays a critical role in preparing learners to understand the natural world, make informed decisions about scientific issues, and contribute meaningfully to sustainable development. As scientific knowledge continues to expand alongside rapid technological advancement, educators are increasingly challenged to adopt teaching approaches that not only promote conceptual understanding but also make learning relevant to learners' cultural, social, and environmental contexts (Clark & Mayer, 2023; Mayer, 2021). Consequently, there has been growing recognition that effective science education should value diverse knowledge systems while making use of innovative instructional approaches that actively engage learners.

In South Africa, curriculum transformation following democracy has sought to create a more inclusive education system that reflects the country's cultural diversity and addresses historical inequalities. One significant outcome of this transformation has been the incorporation of IKS into the national curriculum. The Curriculum and Assessment Policy Statement (CAPS) recognizes Indigenous Knowledge as an important component of Life Sciences education and encourages teachers to integrate indigenous perspectives with scientific concepts to provide learners with a more holistic understanding of biological phenomena (Department of Basic Education [DBE], 2011). Rather than positioning Indigenous Knowledge and scientific knowledge as opposing systems, the curriculum promotes their complementary use to enrich science learning and encourage learners to appreciate multiple ways of understanding the natural world.

Indigenous Knowledge Systems represent the accumulated knowledge, skills, values, and practices developed by Indigenous communities through long-term interaction with their social and natural environments (Aikenhead & Michell, 2011). In the South African context, this knowledge includes traditional understandings of biodiversity, medicinal plants, environmental conservation, agriculture, nutrition, and sustainable resource management. Integrating these knowledge systems into Life Sciences provides opportunities for learners to connect classroom learning with their lived experiences while recognizing the scientific contributions embedded within Indigenous practices (Blackie, 2024; le Grange, 2015). Such integration also supports culturally responsive education by validating learners' cultural identities and promoting greater inclusivity within science classrooms.

Despite the recognised value of Indigenous Knowledge Systems, research consistently indicates that their integration into science education remains challenging. Studies conducted in South African schools have found that although teachers generally acknowledge the importance of IKS, many lack the pedagogical knowledge, confidence, and instructional resources required for meaningful implementation (Mkhwebane, 2024; Photo & McKnight, 2024; Patel et al., 2026). Similarly, a recent systematic review by Ogegbo and Ramnarain (2025) concluded that successful integration requires contextualized instructional materials, learner-centered pedagogies, and stronger collaboration with Indigenous knowledge holders. These findings suggest that while curriculum policy has established the importance of IKS, practical implementation continues to present significant challenges for teachers.

At the same time, multimedia has emerged as an important instructional approach capable of enhancing teaching and learning across science education. Multimedia integrates text, graphics, audio, video, animation, simulations, and interactive activities to present information through multiple representations, thereby supporting learner engagement and conceptual understanding (Clark & Mayer, 2023; Mayer, 2021). Research has demonstrated that well-designed multimedia can improve learners' motivation, facilitate the visualization of complex biological concepts, and promote active participation in learning (Koseoglu & Efendioglu, 2015; Mayer, 2021). These affordances suggest that multimedia may also provide an effective means of supporting the teaching of Indigenous Knowledge by representing oral traditions, cultural practices, ecological knowledge, and community experiences in ways that conventional text-based resources cannot.

Although studies have examined Indigenous Knowledge integration, science education, multimedia learning, and educational technology independently, comparatively little attention has been given to how multimedia can be intentionally designed to support the integration of Indigenous Knowledge Systems within South African Life Sciences education. Existing research has largely focused on curriculum implementation, teacher perceptions, or pedagogical approaches, while the role of multimedia as a curriculum support tool remains underexplored. Consequently, there is a need to synthesize existing literature to determine how multimedia learning principles and educational technologies can support culturally responsive science teaching and strengthen the implementation of Indigenous Knowledge within the Life Sciences curriculum.

Therefore, the purpose of this literature review is to establish the need for multimedia to support the integration of Indigenous Knowledge Systems in South African Life Sciences education. Specifically, the review examines literature relating to Indigenous Knowledge integration, multimedia in science education, learning theories, and multimedia learning principles to identify current challenges, opportunities, and implications for instructional design. In doing so, the review seeks to provide evidence that can inform the development of effective multimedia resources capable of supporting teachers, enhancing learner engagement, and promoting meaningful integration of Indigenous Knowledge within South African Life Sciences classrooms.

2. Indigenous Knowledge Systems in South African Life Sciences Education

The integration of IKS into South African education forms part of the broader curriculum transformation agenda aimed at promoting educational equity, cultural inclusivity, and social justice. Following the introduction of the CAPS, Indigenous Knowledge was formally recognised as an important component of Life Sciences education, reflecting a growing acknowledgement that scientific knowledge is not the only legitimate way of understanding the natural world (Department of Basic Education [DBE], 2011). Instead, the curriculum encourages teachers to integrate Indigenous Knowledge alongside scientific knowledge to provide learners with opportunities to appreciate diverse knowledge systems while developing a deeper understanding of biological concepts.

IKS encompass the accumulated knowledge, beliefs, values, skills, and practices that Indigenous communities have developed over generations through continuous interaction with their environments (Aikenhead & Michell, 2011). In South Africa, this knowledge includes traditional understandings of medicinal plants, biodiversity conservation, food production, ecological sustainability, weather prediction, and natural resource management. Unlike scientific knowledge, which is often generated through controlled experimentation, Indigenous Knowledge is rooted in observation, lived experience, oral traditions, and community practices. Contemporary science education increasingly recognizes that these knowledge systems should not be viewed as mutually exclusive but rather as complementary ways of understanding the natural world (Blackie, 2024; le Grange, 2015).

The educational value of integrating Indigenous Knowledge into Life Sciences has been widely acknowledged within the literature. By incorporating culturally relevant examples into science teaching, learners can connect abstract biological concepts with familiar experiences from their own communities. Such connections promote meaningful learning, increase learner participation, and contribute to culturally responsive teaching practices that recognize learners' identities and lived experiences as valuable educational resources (Blackie, 2024; Mkhwebane, 2024). Furthermore, integrating Indigenous Knowledge encourages learners to critically examine similarities and differences between Indigenous explanations and scientific explanations, thereby fostering critical thinking and scientific literacy rather than viewing one knowledge system as superior to the other (Aikenhead & Michell, 2011).

Within the South African Life Sciences curriculum, Indigenous Knowledge can be meaningfully incorporated into several curriculum topics. Concepts relating to biodiversity, ecology, conservation, nutrition, medicinal plants, and sustainable resource management naturally lend themselves to the integration of Indigenous perspectives because they reflect knowledge that has been developed through generations of interaction with local ecosystems (DBE, 2011). For example, the study of indigenous medicinal plants provides opportunities for learners to explore traditional healing practices while simultaneously examining plant anatomy, physiology, secondary metabolites, and scientific investigations into medicinal efficacy. Such learning experiences enable learners to appreciate the complementary contributions of Indigenous and scientific knowledge to understanding biological phenomena.

Despite these educational benefits, the implementation of Indigenous Knowledge within Life Sciences classrooms remains inconsistent. Recent empirical studies indicate that many teachers continue to experience difficulties integrating IKS because they have received limited professional preparation and often lack confidence in selecting appropriate pedagogical approaches (Mkhwebane, 2024; Photo & McKnight, 2024; Patel et al., 2026). In addition, teachers report having limited access to curriculum-aligned instructional resources that demonstrate practical ways of incorporating Indigenous Knowledge into everyday classroom practice. Consequently, Indigenous Knowledge is frequently presented as isolated examples or omitted altogether, reducing its educational value and limiting opportunities for learners to engage meaningfully with Indigenous perspectives.

Recent reviews further suggest that effective integration requires more than curriculum policy alone. Ogegbo and Ramnarain (2025) found that successful implementation depends on learner-centered pedagogies, contextualized instructional materials, experiential learning opportunities, and meaningful engagement with Indigenous knowledge holders. Similarly, studies examining the representation of Indigenous Knowledge within the Further Education and Training Life Sciences curriculum have argued that stronger instructional support is required if teachers are to implement curriculum expectations consistently and effectively (Ahanonye et al., 2026). These findings indicate that while the policy environment strongly supports the inclusion of Indigenous Knowledge, significant gaps remain between curriculum intentions and classroom implementation.

Collectively, the literature demonstrates that Indigenous Knowledge Systems have an important role to play in promoting culturally responsive and contextually relevant Life Sciences education in South Africa. However, the successful integration of IKS depends not only on curriculum policy but also on the availability of effective pedagogical approaches, instructional resources, and teacher support. Addressing these challenges requires innovative educational solutions capable of presenting Indigenous Knowledge in authentic, engaging, and pedagogically meaningful ways. One promising approach that has gained increasing attention is the use of multimedia to support teaching and learning, which is explored in the following section.

3. The Need for Multimedia to Support the Integration of Indigenous Knowledge Systems

The successful integration of IKS into Life Sciences education depends not only on curriculum reform but also on the availability of instructional approaches that enable teachers to present Indigenous Knowledge in authentic, engaging, and meaningful ways. While policy documents encourage the inclusion of Indigenous perspectives in science education, research suggests that teachers often struggle to move beyond traditional lecture-based instruction due to limited instructional resources and uncertainty about effective teaching strategies (Mkhwebane, 2024; Ogegbo & Ramnarain, 2025). These challenges highlight the need for innovative teaching approaches capable of bridging curriculum expectations and classroom practice.

One instructional approach that has gained considerable attention across education is the use of multimedia. Multimedia refers to the integration of multiple forms of communication, including text, graphics, audio, video, animation, simulations, and interactive elements, within a single learning environment to facilitate teaching and learning (Clark & Mayer, 2023; Mayer, 2021). Unlike conventional text-based instruction, multimedia enables learners to engage with information through multiple representations, thereby creating richer and more interactive learning experiences. Advances in digital technology have further expanded opportunities for multimedia to support inquiry-based learning, collaboration, and learner-centered instruction across a range of educational contexts.

The educational value of multimedia has been widely documented within science education. Biological concepts are often abstract, dynamic, and difficult for learners to visualize using static textbook illustrations alone. Processes such as photosynthesis, cellular respiration, genetics, ecological interactions, and physiological systems involve complex relationships that benefit from visualization through animations, simulations, videos, and interactive models. Research has shown that appropriately designed multimedia can improve conceptual understanding, learner motivation, knowledge retention, and scientific reasoning by helping learners visualize concepts that are otherwise difficult to observe directly (Clark & Mayer, 2023; Koseoglu & Efendioglu, 2015; Mayer, 2021).

The characteristics of Indigenous Knowledge similarly make it well suited to multimedia-supported instruction. Indigenous Knowledge is traditionally transmitted through storytelling, observation, demonstration, participation, and oral communication within communities. These forms of knowledge are often difficult to represent effectively through text alone. Multimedia technologies provide opportunities to preserve and communicate Indigenous Knowledge using digital storytelling, documentary videos, audio recordings, interactive maps, photographs, animations, and virtual field experiences. Such representations allow learners to engage with authentic cultural practices while preserving the social and environmental contexts in which Indigenous Knowledge is developed and applied (Naidoo et al., 2025; Seleke, 2026).

Multimedia also creates opportunities to promote meaningful dialogue between Indigenous Knowledge and scientific knowledge. Rather than presenting Indigenous Knowledge as isolated

examples within the curriculum, interactive multimedia can encourage learners to investigate multiple perspectives, compare explanations, analyze evidence, and reflect critically on how different knowledge systems contribute to understanding biological phenomena. For example, learners studying indigenous medicinal plants could simultaneously explore traditional uses, plant identification, active chemical compounds, conservation concerns, and current scientific research within a single interactive learning environment. Such approaches support inquiry-based learning while encouraging learners to appreciate both Indigenous and scientific perspectives.

The increasing availability of digital technologies has further expanded opportunities for integrating Indigenous Knowledge into science education. Interactive whiteboards, online learning environments, mobile applications, virtual museums, and digital archives provide new platforms through which Indigenous Knowledge can be documented, shared, and explored within formal education. Studies have demonstrated that technology-enhanced learning environments can strengthen learner engagement while providing teachers with flexible instructional resources that support culturally responsive teaching (Mkhwebane et al., 2025; Photo, 2026). Furthermore, digital technologies provide opportunities to preserve Indigenous Knowledge for future generations while ensuring that learners have access to authentic cultural resources that might otherwise be unavailable within the classroom.

Despite these opportunities, research specifically examining multimedia to support the integration of Indigenous Knowledge in South African Life Sciences education remains limited. Existing studies have predominantly focused on curriculum implementation, teacher perceptions, or pedagogical approaches, with comparatively little attention given to the design and evaluation of multimedia resources that support Indigenous Knowledge integration. Consequently, there remains a significant gap in understanding how multimedia learning principles can be applied to develop culturally responsive instructional resources that align with the Life Sciences curriculum. Addressing this gap has the potential to support teachers, enhance learner engagement, and strengthen the meaningful integration of Indigenous Knowledge within South African science education.

4. Learning Theories and Multimedia Learning Principles

The effectiveness of multimedia in education depends not simply on the use of technology but on how multimedia is designed to support learning. Research has consistently shown that learners do not automatically learn more because technology is incorporated into instruction; rather, learning improves when multimedia is grounded in established learning theories and evidence-based instructional design principles (Clark & Mayer, 2023; Mayer, 2021). Within the context of IKS, these theories provide valuable guidance for designing multimedia resources that encourage meaningful engagement, promote conceptual understanding, and support culturally responsive teaching.

One of the most influential theories underpinning multimedia learning is constructivism. Constructivist learning theory proposes that learners actively construct knowledge by connecting new information to their existing knowledge, experiences, and beliefs rather than passively receiving information from teachers (Bruner, 1966; Piaget, 1972). Learning therefore becomes a process of inquiry, interpretation, and reflection in which learners actively make sense of new experiences. This perspective aligns closely with the integration of Indigenous Knowledge because learners often enter the classroom with cultural experiences, community practices, and local environmental knowledge that can serve as valuable foundations for learning scientific concepts. Multimedia resources that encourage exploration, investigation, and learner interaction therefore provide opportunities for learners to build meaningful connections between Indigenous Knowledge and scientific knowledge.

Situated cognition further strengthens the argument for multimedia-supported learning. According to this perspective, knowledge is best acquired within authentic social, cultural, and environmental contexts rather than through isolated facts presented independently of real-life experiences (Brown et al., 1989; Collins et al., 1989). Indigenous Knowledge itself is inherently situated, having developed through generations of observation, participation, and interaction with local environments. Consequently, multimedia resources that incorporate authentic videos, photographs, interviews, digital storytelling, and virtual field experiences can recreate these meaningful contexts and enable learners to experience Indigenous Knowledge in ways that reflect

its traditional modes of transmission. Such learning environments help learners appreciate not only scientific concepts but also the cultural contexts in which knowledge is produced and applied.

Metacognition also plays an important role in multimedia learning because it encourages learners to monitor, regulate, and evaluate their own understanding throughout the learning process (Flavell, 1979). Interactive multimedia environments can promote metacognitive thinking by allowing learners to revisit information, receive immediate feedback, complete reflective activities, and monitor their own progress. These opportunities are particularly valuable when learners are required to compare Indigenous explanations with scientific explanations, evaluate evidence from different knowledge systems, and justify their own conclusions. Supporting learners' metacognitive development encourages deeper understanding while fostering critical thinking and independent learning.

Among the most influential frameworks for multimedia design is Mayer's (2021) Cognitive Theory of Multimedia Learning (CTML). The theory proposes that meaningful learning occurs when learners actively select relevant information, organize it into coherent mental representations, and integrate new knowledge with prior understanding. Mayer further argues that learners process information through separate visual and auditory channels, each with limited capacity, making instructional design critical for preventing cognitive overload while promoting meaningful learning (Mayer, 2021). This theory has become one of the most widely applied frameworks for designing educational multimedia because it provides practical principles for improving instructional effectiveness.

Several multimedia learning principles derived from CTML are particularly relevant to the design of multimedia resources for Indigenous Knowledge Systems. The coherence principle recommends eliminating unnecessary words, images, and sounds that may distract learners from essential information. The signaling principle suggests highlighting key concepts through headings, color, arrows, or other visual cues that direct learners' attention to important information. The segmenting principle recommends dividing complex content into manageable sections, allowing learners to process information at an appropriate pace. The multimedia principle demonstrates that learners understand concepts more effectively when words are combined with relevant visuals than when text is presented alone. Finally, the personalization principle suggests

presenting information using conversational language that promotes learner engagement and creates a more meaningful learning experience (Clark & Mayer, 2023; Mayer, 2021).

Another theoretical perspective that complements multimedia learning is Cognitive Load Theory, which proposes that instructional materials should be designed to reduce unnecessary mental effort and optimize learners' limited working memory capacity (Sweller et al., 2011). Complex scientific concepts, particularly those requiring learners to understand relationships between Indigenous and scientific knowledge, can place considerable demands on working memory if poorly designed instructional materials are used. Multimedia that follows evidence-based design principles can minimize extraneous cognitive load while directing learners' attention toward essential concepts, thereby improving comprehension and long-term retention.

Taken together, constructivism, situated cognition, metacognition, Cognitive Load Theory, and the CTML provide a strong theoretical foundation for the development of multimedia resources that support the integration of Indigenous Knowledge Systems within South African Life Sciences education. Collectively, these theories emphasize that effective multimedia should actively engage learners, connect learning to authentic cultural contexts, support knowledge construction, and promote meaningful interaction with content rather than simply presenting information through digital technologies. Such theoretically informed multimedia has the potential to strengthen culturally responsive teaching while improving learners' conceptual understanding of both Indigenous and scientific knowledge systems.

5. Establishing the Need for Multimedia to Support the Integration of Indigenous Knowledge Systems

The literature reviewed demonstrates that although the integration of IKS has been formally recognised within the South African Life Sciences curriculum, meaningful implementation remains a significant challenge. The inclusion of IKS within the CAPS reflects an important shift towards culturally responsive education; however, policy alone is insufficient to ensure successful classroom implementation (DBE, 2011). Across the literature, researchers consistently argue that effective integration requires appropriate instructional resources, pedagogical support, teacher professional development, and teaching approaches that enable learners to engage meaningfully

with both Indigenous and scientific knowledge (Ahanonye et al., 2026; Mkhwebane, 2024; Ogegbo & Ramnarain, 2025).

One of the most frequently reported challenges concerns teacher preparedness. Although many teachers recognize the educational value of Indigenous Knowledge, they often report limited confidence in incorporating it into science teaching because they have received little formal preparation during their pre-service education or subsequent professional development (Mkhwebane, 2024; Patel et al., 2026; Photo & McKnight, 2024). Teachers also indicate uncertainty regarding appropriate pedagogical approaches, suitable learning activities, and methods for balancing Indigenous perspectives with scientific explanations. Consequently, Indigenous Knowledge is frequently presented as isolated examples rather than being meaningfully integrated throughout Life Sciences lessons.

The literature further highlights a shortage of curriculum-aligned instructional resources that support the teaching of Indigenous Knowledge. Existing textbooks often provide only brief references to Indigenous practices without offering opportunities for learners to investigate, compare, or apply Indigenous and scientific knowledge within authentic learning contexts (Ahanonye et al., 2026). As a result, teachers are frequently required to develop their own materials, a process that can be time-consuming and challenging, particularly for those with limited experience in integrating IKS. This lack of accessible teaching resources represents one of the major barriers to successful curriculum implementation.

Another recurring theme within the literature is the importance of learner engagement. Contemporary science education increasingly emphasizes active participation, inquiry-based learning, collaboration, and authentic problem-solving rather than passive transmission of knowledge (Clark & Mayer, 2023; Mayer, 2021). However, traditional teaching approaches often limit learners' opportunities to interact with Indigenous Knowledge in meaningful ways. Instead of exploring real-world contexts, learners may encounter Indigenous Knowledge only through brief textual descriptions or teacher explanations, reducing opportunities to appreciate its relevance, complexity, and practical applications. These findings suggest that more interactive instructional approaches are needed to support meaningful learning.

Multimedia offers considerable potential to address many of these challenges. By combining text, images, audio, video, animation, simulations, and interactive activities within a single learning environment, multimedia enables learners to engage with concepts through multiple representations while supporting different learning preferences (Clark & Mayer, 2023; Mayer, 2021). Within the context of Indigenous Knowledge, multimedia provides opportunities to present oral histories, traditional ecological knowledge, indigenous medicinal plants, cultural practices, and community experiences using authentic visual and auditory resources that cannot easily be conveyed through printed text alone. Such learning experiences may strengthen learner engagement while simultaneously promoting cultural appreciation and scientific understanding.

The literature also suggests that multimedia can serve as an important source of professional support for teachers. Digital learning resources can provide structured lesson materials, interactive activities, assessment opportunities, videos, and virtual demonstrations that assist teachers in implementing curriculum requirements with greater confidence. Furthermore, multimedia resources developed using evidence-based instructional design principles can reduce teachers' planning demands while ensuring that learning activities remain aligned with curriculum objectives and effective pedagogical practice (Clark & Mayer, 2023; Mkhwebane et al., 2025).

An additional consideration emerging from the literature is the importance of preserving and documenting Indigenous Knowledge. Much Indigenous Knowledge continues to be transmitted orally within communities, creating concerns regarding its long-term preservation as knowledge holders age and societal practices change. Digital technologies and multimedia therefore offer opportunities not only for improving classroom instruction but also for documenting Indigenous languages, cultural practices, medicinal plant knowledge, and environmental management strategies for future generations (Naidoo et al., 2025; Seleke, 2026). When developed collaboratively with Indigenous communities and knowledge holders, multimedia resources can contribute to both educational improvement and cultural preservation.

The literature establishes a clear educational need for multimedia that supports the integration of Indigenous Knowledge Systems within South African Life Sciences education. While curriculum policy provides the foundation for integration, successful implementation requires instructional resources that are culturally responsive, pedagogically sound, and grounded in established

multimedia learning principles. Developing such resources has the potential to support teachers, enhance learner engagement, promote meaningful connections between Indigenous and scientific knowledge, and contribute to a more inclusive and contextually relevant science education system.

6. Conclusion

The integration of IKS into South African Life Sciences education represents an important step towards creating a science curriculum that is culturally responsive, socially relevant, and reflective of the country's diverse knowledge traditions. The literature reviewed demonstrates that Indigenous Knowledge has considerable educational value because it enables learners to connect scientific concepts with local experiences, cultural practices, and environmental knowledge. However, despite strong curriculum support through CAPS, meaningful implementation remains constrained by challenges relating to teacher preparedness, limited instructional resources, and uncertainty regarding effective pedagogical approaches.

The review also highlights the significant potential of multimedia to address many of these challenges. Research indicates that multimedia, when designed according to established learning theories and multimedia learning principles, can enhance learner engagement, improve conceptual understanding, support inquiry-based learning, and create authentic learning experiences. Within the context of Indigenous Knowledge Systems, multimedia provides unique opportunities to represent oral traditions, cultural practices, ecological knowledge, and community experiences through interactive and visually rich learning environments that are difficult to achieve using conventional teaching resources alone.

Theoretical perspectives such as constructivism, situated cognition, metacognition, Cognitive Load Theory, and Mayer's Cognitive Theory of Multimedia Learning further demonstrate that effective multimedia should be intentionally designed to promote active knowledge construction, contextual learning, learner reflection, and efficient cognitive processing. These theories collectively provide a strong instructional design foundation for developing multimedia resources that support both Indigenous and scientific ways of knowing.

This literature review establishes a clear need for the development of multimedia resources that support the integration of Indigenous Knowledge Systems in South African Life Sciences

education. Such resources have the potential to strengthen curriculum implementation, support teachers in delivering culturally responsive instruction, enhance learner engagement, and promote deeper understanding of biological concepts through meaningful connections between Indigenous Knowledge and science. Future research should therefore focus on designing, implementing, and evaluating multimedia resources in collaboration with teachers, curriculum specialists, Indigenous knowledge holders, and local communities to ensure that these resources are educationally effective, culturally authentic, and responsive to the needs of South African classrooms.

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